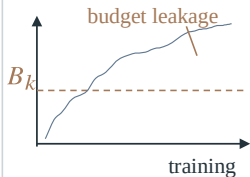


Penalty view

Costs enter the objective

Fixed penalty



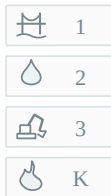
Fixed prices can leave budget leakage.

(schematic)

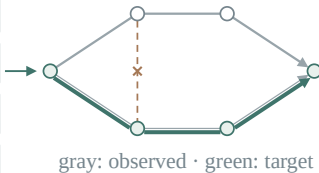
SafeOT

One flow, all budgets

Soft costs



Entropic projection



gray: observed · green: target

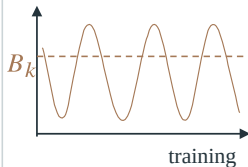
× hard constraints remove edges

Safety prices before violations

Constraint view

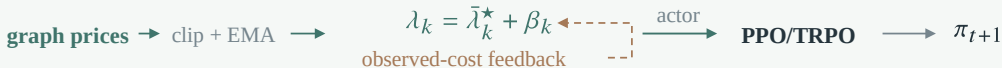
Budgets bound costs

Lagrangian feedback



Observed-cost feedback can lag and oscillate.

(schematic)



One price rule

Fixed penalty

$c = 0, \beta_k$ fixed

exact settings

Lagrangian

$c = 0, \beta_k$ adaptive

SafeOT

finite ε

LP

$\varepsilon \rightarrow 0$

conditional limits

Active-set

$\varepsilon \rightarrow \infty$, saturated clip